MILITARY SPECIFICATION

SCALER, NEEDLE TYPE, PNEUMATIC, PORTABLE

This amendment forms a part of Military Specification MIL-S-23492B, dated 3 August 1973, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

1.2, Delete "Size 1 - Scaler, 6 pounds, 19 inches." and substitute "Size 1 - Standard duty"; and delete "Size 2 - Scaler, 8.5 pounds, 20 inches." and substitute "Size 2 - Heavy duty".

Page 2

2.2, Delete all reference to "DEPARTMENT OF LABOR" document.

Page 3

3.9, second line, Delete "style" and substitute "model number".

Page 4

3.11, Delete and substitute:

"3.11 Throttle. The throttle shall be manually operated and shall automatically and positively shut off the air supply to the tool when released. The throttle trigger or lever shall be located on the handle and shall insure ease of operation under service conditions. The throttle valve shall be fitted with "O" rings or gaskets and shall prevent leakage of air greater than the amount specified in 4.6.4 when in the closed position."

3.12, Delete and substitute:

"3.12 Needles. A minimum of 10 needles shall be provided. Needles shall have a minimum length of 6 inches, shall be installed substantially parallel to housing, and shall be capable of having a 2-1/2-inch extension beyond the nose end of the scaler. The needle hardness shall conform to table I. The needle body diameter shall be 0.120 ± 0.006 inch, the needle head diameter shall be 0.190 ± 0.015 inch, and the needle head height shall be 3/32 ± 1/32 inch."

3.19.1, second line, Delete "90 decibels" and substitute "90 decibels".

3.20, Delete in its entirety and substitute:

"3.20 Workmanship. Workmanship of the scalers shall be such that the tools shall be free of defects which would adversely affect the operation, efficiency, serviceability, and life expectancy of the tools. All components shall be new and machined to tolerances which insure ease of interchangeability."

Page 5

4.5.1, Delete after colon and substitute:

"A steel base plate 1-3/16 inch thick, 4-1/2 inches wide, and 34 inches long. Two split holding blocks 1/2 by 2-3/4 by 4 inches are attached to the base and are used to hold two oil-lite bushings. Oil-lite bushings shall be machined to suit needle scaler housing with 0.005-inch clearance for free axial movement with a spring at end of scaler to provide cycling loading on the scaler. A striking plate made of 4-inch angle plate is attached to the base plate, as is the striking plate locking device. A test specimen, against which the needles will impact, is backed by rubber and attached to the striking plate. The test specimen is 3/8-inch thick steel, 4 inches by 4 inches and in accordance with MIL-S-20166. The rubber backing is 1/2 inch thick and also 4 inches by 4 inches and has a Shore "A" Durometer hardness of 60 to 70."

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4.5.2, third line, after "plate", delete remainder of sentence.

Page 6

4.6.4, Delete in its entirety and substitute:

"4.6.4 Air leakage test. Scalers shall be tested for tightness. Scalers shall be connected to a system consisting of a 0.25 cubic foot air receiver, valving as necessary and a 0 to 300 lb/in^2 range pressure gage. System shall be pressurized with air or nitrogen to a gage pressure of 150 lb/in^2 with throttle in the closed position and isolated for a period of 15 minutes. At the end of 15 minutes, maximum pressure drop permitted shall be no more than 50 lb/in^2."

4.6.6.1, Delete in its entirety.

4.6.6.2 and 4.6.6.3, Delete in their entirety and substitute:

"4.6.6.2 Irregular surface test plate. Irregular surface test plate shall be a single "V" weld joint assembly which is attached to a striking plate with screws. The test plate shall be made of 0.375-inch thick plate, double beveled 30 degree angles with a 0.25-inch root opening at the backing plate. A piece of stiff cardboard shall be placed between the backing plate and the beveled plates.

"4.6.6.3 Procedure. The test scaler shall be hand held and operated for one minute. During this time, the operator shall maneuver the scaler to varying positions about the irregular surface test plate. After completion of test, the cardboard shall be examined for indication that the needles have made contact with the backing plate. Failure of making contact with the backing plate shall be cause for rejection."

Page 7

4.6.7.3, Delete and substitute:

"4.6.7.3 Procedure. The needles and spring of the needle scaler shall be replaced with a chisel blank 7.5 inches long with a squared end. The modified scaler shall then be locked into the carriage of the testing machine at an angle of 15 degrees to the horizontal. Size 1 scalers shall be pulled by a 25-pound load applied in a horizontal direction on the carriage. Size 2 scalers shall be pulled by a 50-pound load. The scaler shall be operated with the throttle fully open. The dimensions of the cut shall be: width - 3/16 inch, length - 20 inches, and depth - 0.035 ± 0.005 inch for size 1 and 0.050 ± 0.005 inch for size 2. The time required to make a 20-inch long cut shall be recorded. The chip removed shall be weighed and the number of grams of metal removed per minute calculated. This test shall be conducted three times and the average of the results of the tests shall be used to determine conformance to the requirements of 3.17 and table I."

4.7, *Add at end of paragraph:*

"When test under practical service conditions is not feasible, a simulated test may be performed of equal duration by using a life testing machine. Tool shall be placed in test apparatus as shown in figure 5 and cycled on for 20 seconds and off for 10 seconds."

Page 8

Add the following new paragraph:

"6.6 This specification reflects the requirements of OSHA, in effect as of the date of this specification."

Page 9

Figure 5, Delete and substitute the attached new figure 5.

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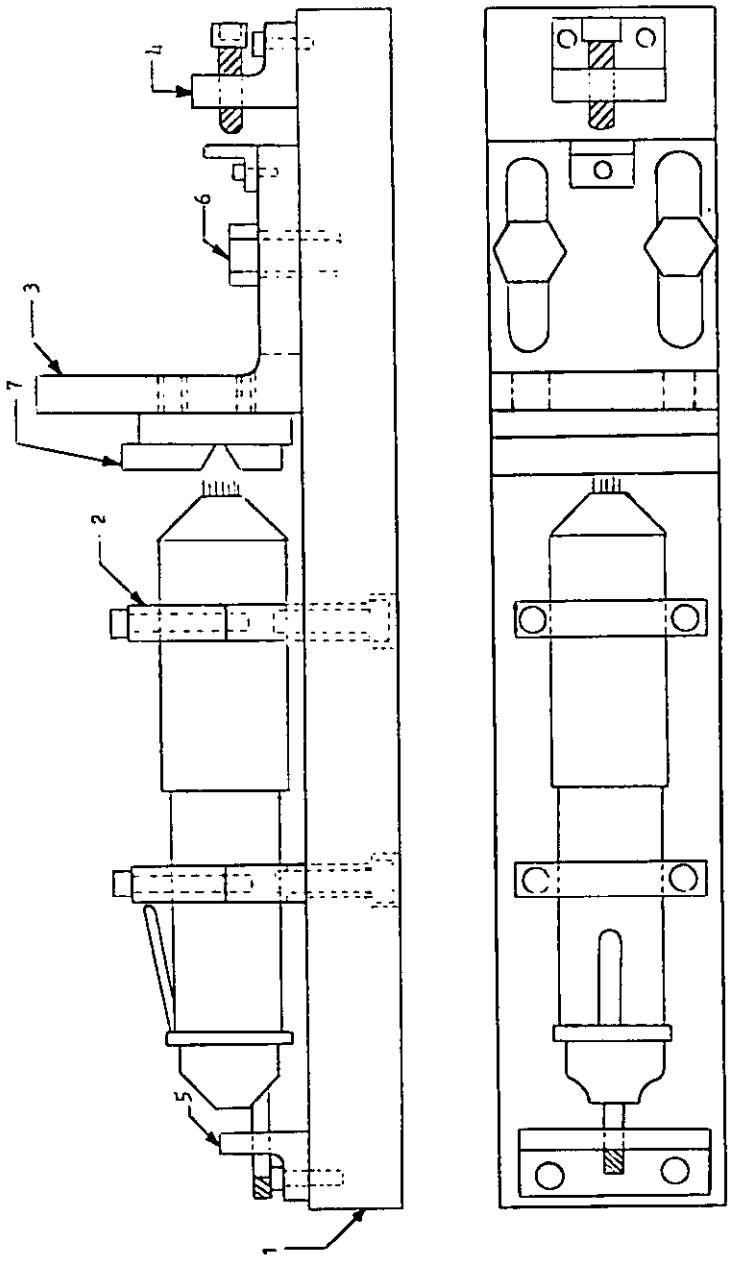
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CHL0540

Figure 5 - Endurance and service tests fixture with specimen in place.

1. Base plate 1-3/16 by 4-1/2 by 14 inches.
2. Split holding blocks 1/2 by 2-3/4 by 4 inches.
3. Striking plate 4-inch angle plate.
4. Striking plate locking device.
5. Needle sealer locking device.
6. 3/4-inch - 10 bolts.
7. Test specimen assembly.
8. Bushings (oil-lite).
9. Spring.

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